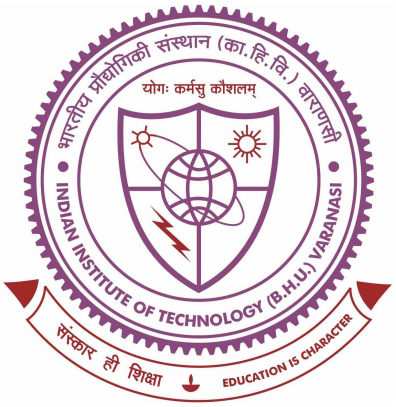
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Dissertation

submitted in partial fulfilment for the Award of

Doctor of Philosophy

by

FULL NAME

under supervision of

PROF. ABC

SCHOOL OF BIOMEDICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY

(BANARAS HINDU UNIVERSITY)

VARANASI 221005

UTTAR PRADESH, INDIA

2002 August, 2026

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It is further certified that the student has fulfilled all the requirements of Comprehensive Examination, Candidacy and SOTA for the award of **Doctor of Philosophy**.

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Signature of the Student

Date: October 15, 2026
Place: Varanasi, India

CERTIFICATE BY THE SUPERVISOR

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Dedicated to my beloved family, whose unwavering love, support, and encouragement have been my constant source of strength throughout this journey.

SYNOPSIS

Name of the student:	**Full Name**
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Dissertation Title:	***Full title of this dissertation***
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July 2026

Organization of the Dissertation

Organization of this dissertation: This dissertation is made up of the following sequence of chapters:

Chapter 1: Introduction and review of literature, Research gap identification, Approach and Main objectives.

Chapter 2: Materials, methods, equipment design, and computational approaches

Chapter 3: Biological experiment results, discussion of obtained results and major findings.

Part 1: Chapter 1

Part 2: Chapter 2

Part 3: Chapter 3

Chapter 4: Conclusions of this dissertation and future aspects of this work.

All the references and bibliography have been included at the end of each of the respective chapters and their parts.

Chapter 1: Introduction and review of literature

Objectives of this dissertation

1. **Objective 1:** 1
2. **Objective 2:** 2
3. **Objective 3:** 3

Chapter 2: Materials and Methods

This chapter in this dissertation discusses about the all the materials and methods used for this study.

Chapter 3: Results and discussion

Part 1

Paper 1

Part 2

Paper 2

Part 3

Paper 3

Chapter 4: Conclusions and future aspects

In this chapter 4, I have summarized and concluded all the results obtained from the work reported in this dissertation. Future scope of the this work has been discussed as a part of this chapter.

Outcomes from this dissertation

1. **Chapter:3 part: 1** Paper 1. referance full
2. **Chapter:3 part: 2** Paper 2. referance full
3. **Chapter:3 part: 3** Paper 3. referance full

Extended Abstract

Extended Abstract

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List of Abbreviations

Abbreviation	Full Form
ABH	(α-amino)boronophenylalanine (arginase inhibitor)
ADC	Analog-to-Digital Converter
AIBN	Azobisisobutyronitrile

List of Symbols

Symbol	Meaning
α	Alpha;
β	Beta;
γ	Gamma;
κ	Kappa;
μ	Micro (prefix, 10^{-6}); e.g. μg, μL, μM
$\pm$	Plus or minus; indicates standard deviation or uncertainty
$\pi \rightarrow \pi^*$	Pi-to-pi-star electronic transition (UV-Vis absorption)
v	Reaction rate (enzyme kinetics)

(Continued on next page)

(Continued from previous page)

Symbol	Meaning
%	Percentage
°**C**	Degrees Celsius
nm	Nanometre (10^{-9} m); particle size unit
nM	Nanomolar (10^{-9} mol L^{-1})
μ**M**	Micromolar (10^{-6} mol L^{-1})
mg mL^{-1}	Milligrams per millilitre; concentration unit
μ**g mL**$^{-1}$	Micrograms per millilitre; concentration unit
mV	Millivolt; ECG amplitude unit
Hz	Hertz; frequency unit
kDa	Kilodalton; molecular weight unit
eV	Electron volt; XPS binding energy unit
C=O	Carbonyl functional group
NH$_2$	Amine functional group
OH	Hydroxyl functional group
Mn(II)	Manganese(II) ion (catalytic centre of arginase)
n	Sample size
p	p-value (statistical significance)
r^2	Coefficient of determination

Chapter 1

Introduction, literature review, research gap and objective

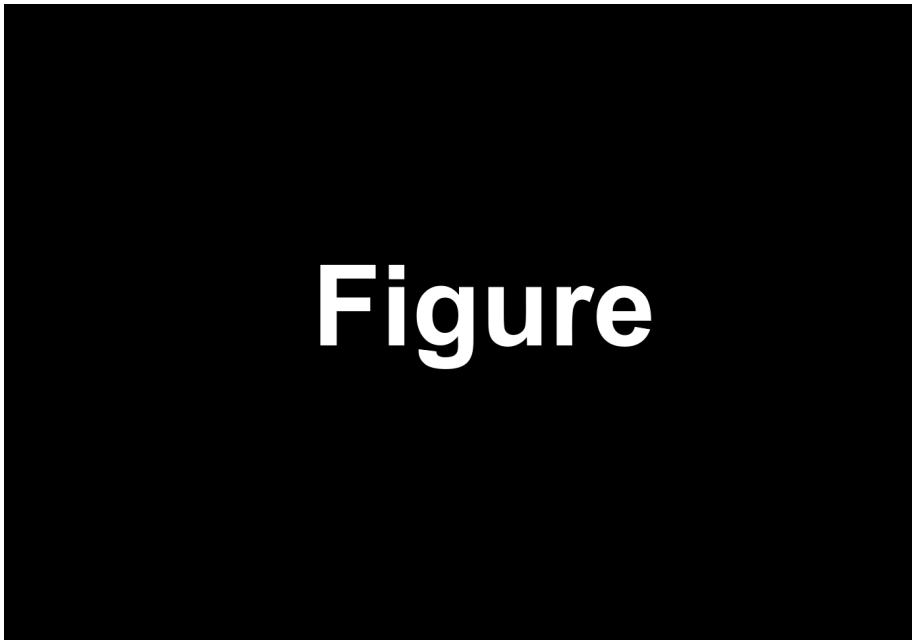


Figure 1.1: Different pathways involved in immunosuppressive microenvironment through MDSCs recruitment and TAMs secreting TGF-β, CCL-22, PD-L1, and IL-10 to promote T-regulatory function and arginine depletion. Tumor immune evasion through PD-1/PD-L1 and CTLA-4 checkpoint activation, IDO-mediated tryptophan catabolism, and adenosine pathway signaling via CD39/CD73, collectively suppressing CTL and NK cell activity. These mechanisms converge to downregulate MHC-I expression, reduce DC antigen presentation, and inhibit cytotoxic T lymphocyte activation, differentiation, and proliferation.

1.1 Main objectives of this dissertation

On the basis of the above discussion about the review of literature, related definition of the problem, contemporary challenges, and drawbacks of existing therapies along with the need for new improvement. Following main objectives of this dissertation are elucidated

Table 1.1: Different types of Quantum dots, their origin, synthesis and biomedical applications

S.No.	Type	Origin	Synthesis	Biomedical applications	Ref.
1.	Carbon quantum dots (CQDs)	CQDs derived from carbon precursors; often grouped with diamond dots	Hydrothermal, microwave, electro chemical, laser, and pyrolysis methods	Bioimaging, biosensing, drug delivery, and theranostics	1
2.	Graphene quantum dots (GQDs)	Graphene derived carbon nanomaterials	Top-down cutting of graphene/ graphite materials or bottom-up synthesis using hydrothermal, microwave, and electro chemical routes	Bioimaging, drug delivery, biosensing, tissue engineering, antibacterial/ antiviral uses, and theranostics	1
3.	InP QDs	Indium phosphide based semiconductor QDs	Synthesized mainly by colloidal methods and core/shell growth with surface passivation	Biomedical fluorescence imaging, immunoassays, biosensing, drug delivery, and diagnostics	1
4.	Cd-based QDs	CdSe and CdTe	Organo metallic, water-based, or microbial based bottom up synthesis; core/shell designs to improve stability and reduce toxicity	Imaging, diagnostics, and other biomedical uses	1
5.	Cd-free quantum dots	InP based, silicon, and other cadmium free systems	Diverse synthesis methods	Biophotonics, nanomedicine, bioimaging, sensing, drug/gene delivery, and theranostics	1
6.	Silicon quantum dots (SiQDs)	Silicon	Laser ablation and other synthesis approaches	In vivo imaging, drug delivery, and tissue engineering	1

below in brief.

- **Objective 1:** Objective 1.
- **Objective 2:** Objective 2.
- **Objective 3:** Objective 3.

Organization of this dissertation: This dissertation comprises the following sequence of chapters:

Chapter 1: Introduction and review of literature.

Chapter 2: Materials, methods, equipment design, and computational approaches.

Chapter 3: Biological experiment results, discussion of obtained results, and major findings.

Chapter 4: In Chapter 4 of this dissertation, we have summarized the results obtained through this dissertation are concluded. The future scope of this work also has been included at the end of this chapter.

Chapter 2

Materials and Experimental procedure

2.1 Materials

Materials used are mentioned here

2.2 Methods

Methods used are mentioned here

2.3 Statistical Analysis

ANOVA and Statistical analysis was performed using OriginPro software. In order to determine the statistical significance of differences between treatment groups (Control, VEGF, Anti-VEGF, and Hydrocortisone) one-way analysis of variance (ANOVA) was conducted. Tukey's HSD post-hoc test was applied to pairwise comparisons for identification of differences between the groups at the significance level of 0.05. Results are reported as statistically significant, where the significance levels are denoted as follows: ns ($p > 0.05$): Not significant, * ($p \leq 0.05$): Significant, ** ($p \leq 0.01$): Very significant, and *** ($p \leq 0.001$): Highly significant.

Chapter 3

Results and discussion

Part I

An integrated deep learning and Bayesian computational method-based quantification of chick embryo angiogenesis model

Part II

Study the Anti-breast cancer effects of carbon quantum dots: *in vivo* angiogenesis and arginase co-inhibition

Part III

Study the Protective effects of cross-linked poly (N-acryloyl glycine) nanoparticles in Type 2 diabetes-induced complications: *in vivo* study

CHAPTER 3, PART I

Paper 1

Published in *Journal*, Paper

Paper 1 reference

3.1 Abstract

Angiogenesis is a process for the formation of new blood vessels from existing vessels and it sits directly at the intersection of normal physiology and multiple diseases. Normal physiological activities like embryonic development, tissue homoeostasis require precise regulation.

3.1.1 Statistical Analysis

Detailed ANOVA and Statistical analysis method is mentioned in section 2.3 on page 5.

3.2 Results and Discussion

This how to cite figure. Normalization of images redistributes pixel intensities to a consistent dynamic range and helps in suppression of background gradients while vessel contrast is preserved. ROI cropping helps in elimination of peripheral tissue and non-vascular background

CHAPTER 3, PART II

Paper 2

Published in *Journal*, Paper

Paper 2 reference

3.3 Abstract

Abstract is placed here.

3.4 Introduction

Introduction is placed here

3.4.1 Results

Chemical formulas, Units and Greek letters are mentioned as given below. H_2SO_4 C=O, bending sodium Auger NaKLL (536 eV) amine ($-NH_2$) 1080 cm^{-1} TNF-α and IL-1β 100 mM 30 min in an incubator at 37 °C. AIBN ($(CH_3)_2C(CN)N=NC(CN)(CH_3)_2$) divinylbenzene ($C_6H_4(CH=CH_2)_2$) (0.3 g)

CHAPTER 3, PART III

Paper 3

Published in *Journal*, Paper

Paper 3 reference

3.5 Abstract

Abstract is here

3.6 Introduction

Introduction is here

3.7 Results and Discussion

ξ of $-39\,\text{mV}$ ^{1}H-NMR and ^{13}C-NMR ($\lambda = 220\,\text{nm}$ to $240\,\text{nm}$) represents a $\pi \rightarrow \pi^{*}$ transition. $\pi \rightarrow \pi^{*}$ transition of phenyl groups.

$$\text{Cell Viability (\%)} = \left(\frac{A_{\text{treated}} - A_{\text{blank}}}{A_{\text{control}} - A_{\text{blank}}} \right) \times 100 \tag{3.1}$$

Chapter 4

Conclusions and future aspects

Conclusion is here

Appendix starts here

Appendix A

Chapter3 (Part1)

Appendix B

Chapter3 (Part2)

Experimental Procedures

X-Ray Diffraction

Approximately 1 g of powdered sample was placed in a sample holder of the XRD machine and scanned at 0.1 step size; readings were recorded using a Co source ($K_\alpha = 1.79$ Å) at 25 °C from $2\theta = 5°$ to $90°$.

Measurement of Fluorescence Quantum Yield

Powder samples were dispersed in distilled water (1 mL) at room temperature (25 °C) and the sample absorbance was set to 0.1 a.u. at the absorbance maximum ($\lambda_{\text{max}} = 325$ nm). The absorption maximum was then used as the excitation wavelength and the emission spectrum was recorded using a fluorescence spectrophotometer. The relative quantum yield was calculated against coumarin as the reference standard using Equation B.1:

$$\phi_f^i = \frac{F_i \, f_s \, n_i^2}{F_s \, f_i \, n_s^2} \, \phi_f^s \tag{B.1}$$

where ϕ_f^i and ϕ_f^s are the photoluminescence quantum yields of the sample and standard, respectively; F_i and F_s are the integrated fluorescence intensities of the sample and reference; $f_x = 1 - 10^{-A_x}$ is the absorption factor (A is the absorbance of the solution); and n_i and n_s are the refractive indices of the sample and reference, respectively.

Supporting Tables

Table B.1: Genes and proteins co-expressed with ARG1 and their associated functions.

S.No.	Gene	Protein	Function
1	MDK	Midkine (MK)	Growth factor
2	SDC4	Syndecan-4	Cell adhesion
3	LILRB1	Leukocyte Ig-like receptor B1	Inhibitory cytokine receptor
4	IFNA2	Interferon alpha-2	Immunomodulatory cytokine
5	IFNL1	Interferon lambda 1	Antiviral cytokine
6	LILRB4	Leukocyte Ig-like receptor B4	Inhibitory receptor on myeloid cells
7	LILRB2	Leukocyte Ig-like receptor B2	Inhibitory immune receptor
8	LGALS9C	Galectin-9C	Cell–cell interactions
9	IFNB1	Interferon beta-1	Antiviral cytokine
10	HLA-G	Human Leukocyte Antigen G	NK- and T-cell inhibitor
11	HLA-E	Human Leukocyte Antigen E	NK-cell modulator
12	ANXA1	Annexin A1	Leukocyte migration
13	LGALS9	Galectin-9	Immune regulation
14	LGALS9B	Galectin-9B	Immune regulation
15	CTSG	Cathepsin G	Neutrophil protease
16	HMGB1	High Mobility Group Box 1	DAMP protein

Continued on next page

S.No.	Gene	Protein	Function
17	CD86	CD86 (B7-2)	T-cell inhibition
18	PDCD1LG2	PD-L2	PD-1 ligand
19	TNFAIP8L2	TNFα-induced protein 8-like 2	Immune regulation
20	CD80	CD80 (B7-1)	CTLA-4 co-stimulator
21	HAVCR2	TIM-3	Immune checkpoint receptor
22	CD274	PD-L1	PD-1 ligand
23	CR1	Complement receptor 1 (CD35)	Complement regulation
24	CTLA4	CTLA-4	Immune checkpoint receptor
25	FOXP3	Forkhead box P3	T-regulatory cell transcription factor
26	BMP4	Bone Morphogenetic Protein 4	Immune modulation
27	HLA-DRB1	HLA-DR Beta 1	MHC class II
28	FGL1	Fibrinogen-like Protein 1	T-cell suppressor
29	TWSG1	Twisted Gastrulation BMP Modulator 1	Immune differentiation
30	VTCN1	B7-H4	T-cell suppressor
31	LAX1	Lymphocyte Transmembrane Adaptor 1	TCR inhibitor
32	PTPN22	Protein Tyrosine Phosphatase NR 22	TCR-signalling inhibitor

Continued on next page

S.No.	Gene	Protein	Function
33	TIGIT	TIGIT	T-cell inhibitor
34	ARG1	Arginase 1	Arginase
35	VSIR	VISTA	Immune checkpoint receptor
36	VSIG4	CRIg	Complement receptor

Table B.2: Urine analysis of mice from acute toxicity study at a dose of 2000 mg kg^{-1}.

S.No.	Parameter	Normal range	Units	Day 1	Day 8	Day 15	Day 28
1	Colour	Clear slightly yellow	–	Pale yellow	Pale yellow	Straw	Pale yellow
2	Transparency	Transparent	–	Transparent	Transparent	Transparent	Transparent
3	Leukocytes	10	cells/µL	15	15	125	70
4	Nitrite	Absent	–	Absent	Absent	Absent	Absent
5	Urobilinogen	1.68–16	µmol/L	16	16	16	16
6	Protein	1.52	g/L	1.0	1.0	1.0	0.3
7	pH	5.0–5.6	–	6.5	7.5	7.0	7.5
8	Blood	Absent	–	−ve	−ve	−ve	−ve
9	Specific gravity	1.045–1.064	–	1.030	1.030	1.015	1.015
10	Ketone bodies	0.6	mmol/L	−ve	−ve	0.5	0.5
11	Bilirubin	1.71–11.97	µmol/L	100	100	17	50
12	Glucose	0.033–0.1443	mmol/L	>5	>5	>5	−ve

Appendix C

Chapter3 (Part3)

Methods

Appendix D

Documentation